

Artificial Intelligence in Mitigating Security Threats for Lightweight IoT Devices: A Survey of Technologies, Protocols, and Future Challenges

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Abstract—Lightweight Internet of Things (IoT) devices—microcontroller-class nodes with less than 512-kB RAM, sub-100-MHz clocks, and low-power radios [Bluetooth low energy (BLE), Zigbee, LoRa, and narrowband IoT (NB-IoT)]—are now widely deployed in settings, where traditional security stacks are infeasible. This survey examines how artificial intelligence (AI) can harden such constrained platforms against device-, network-, and application-layer threats, including spoofing, routing manipulation, DDoS, malware, and advanced persistent threats (APTs). We: 1) formalize a *lightweight envelope* that bounds feasible defenses in terms of RAM, CPU, bandwidth, and energy; 2) consolidate protocol-side risks across BLE, Zigbee, and LoRaWAN; and 3) review deployable AI techniques through a *deployment-first* lens that separates training [edge, cloud, and federated learning (FL)] from on-device inference. Distinct from prior surveys, we provide resource-annotated comparisons that report accuracy *alongside* model size, peak RAM, latency, and estimated energy per inference, showing how pruning, posttraining quantization, distillation, and feature narrowing shift feasibility on microcontroller unit (MCU) targets. Covered methods include compact classifiers [linear models, trees, and support vector machine (SVM)], quantized TinyCNN/TinyRNN, and graph-based intrusion detection (IDS), reinforcement learning (RL) for adaptive rate limiting and channel selection, and privacy-preserving FL with update compression. We conclude with a pragmatic agenda—energy-adaptive inference, low-power wide-area network (LPWAN)-aware scheduling and FL, robustness to poisoning and evasion, and reproducible benchmarks that couple accuracy with size/latency/energy on real hardware—aimed at making AI-based security practical at scale for lightweight IoT deployments.

Index Terms—Energy efficiency, Internet of Things (IoT), intrusion detection (IDS), lightweight devices, low-power wide-area network (LPWAN) (LoRaWAN/Bluetooth low energy (BLE)/Zigbee), security, TinyML, training versus inference.

I. INTRODUCTION

THE Internet of Things (IoT) has become a transformative force across sectors, connecting billions of physical devices embedded with sensors, actuators, and software

to enable intelligent monitoring, automation, and decision-making. IoT is commonly defined as a system of connected physical objects embedded with sensors, software, and control systems that collect and exchange data over networks [1], [2]. These devices typically offer limited computation, storage, and communication capacities, which amplifies their exposure to cyber and privacy risks. While IoT is tightly woven into contemporary digital society, it also faces significant cybersecurity and data-privacy risks. Examples include DDoS attacks, intrusions, botnets, advanced persistent threats (APTs), and malware [3]. These risks are amplified in lightweight deployments due to minimal onboard security and tight budgets of computation, memory, bandwidth, and battery [4]. When optimized for low-power hardware, AI/ML methods can provide proactive defenses while respecting device energy and compute limits [5], [6].

Authentication is a pivotal mechanism for preserving integrity and confidentiality in lightweight IoT systems [7], [8]. Traditional schemes that demand substantial computation and memory are often impractical, whereas AI-enabled lightweight authentication can learn device/usage patterns and flag irregular access with low overhead [9], [10], [11].

As the frontline of defense, lightweight AI/ML models are critical to safeguarding constrained devices [1], [9]. Compact IDS models can learn normal traffic patterns and detect DDoS activity in real time [12]. Intrusions are identified by models trained on device/usage behavior [13]. Botnets and APTs can be detected by techniques that capture the subtle signatures of command-and-control (C&C) and lateral movement [1], [14], [15]. For malware, AI provides a dynamic defense that adapts to evolving threats [16], [17]. When appropriately compressed or pruned, such models can execute on constrained edge devices, enabling decentralized protection across IoT deployments.

A. Existing Surveys

The literature to date on lightweight IoT remains limited, although several surveys cover different aspects of IoT security. Based on our review, most existing surveys do not focus specifically on AI techniques for *lightweight* IoT. Table I summarizes key security issues in IoT networks along with representative ML techniques applied to address them. For clarity, we organize reviewed works into two tables. Table I compiles surveys spanning security/architecture/services and notes how our work extends them; a companion table (see Table II, introduced later) focuses on ML/DL surveys with

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TABLE I
SURVEYED LITERATURE ON AI-DRIVEN LIGHTWEIGHT IoT SECURITY

Year	Reference	Focus Areas	Sections	Gaps Addressed in Our Work
2018	[4]	ML-based IoT security	Sec. II, III	Lacks lightweight model design and performance benchmarking under constraints.
2020	[18]	AI for edge computing in IoT	Sec. III	Does not evaluate AI models on constrained devices or compare deployment costs.
2021	[7]	Lightweight mutual authentication	Sec. IV	No analysis of adaptive authentication based on AI or model compression.
2021	[17]	AI-enabled blockchain access control	Sec. III	Lacks comparative study of latency and computational trade-offs for edge nodes.
2022	[19]	IoT intrusion detection survey	Sec. IV	Does not cover inference-time profiling on embedded platforms.
2023	[1]	ML against APTs in IoT	Sec. IV	Lacks structured threat modeling and taxonomy for lightweight settings.
2023	[11]	Secure SDN-based edge framework for IoT healthcare	Sec. III	No analysis on runtime adaptation or adversarial resilience for DL models.
2023	[20]	FESA: Feature-space-aware IoT IDS	Sec. IV	Lacks training–inference complexity comparison or real-world deployment validation.
2024	[21]	Deep reinforcement learning for IoT traffic anomaly detection	Sec. V	Requires clearer profiling of energy–accuracy trade-offs in lightweight agents.
2024	[22]	Federated learning for IoT malware detection	Sec. IV, V	Omits update compression and scheduling for low-bandwidth nodes.
2025	[23]	Knowledge distillation for edge AI	Sec. IV	Focuses on vision tasks—does not extend to lightweight security models or schedule-aware trade-offs.
2025	This Work	AI for lightweight IoT security	All	Layered comparison; training vs. inference complexity and scheduling on constrained IoT.

TABLE II

COMPARISON OF COMMON LIGHTWEIGHT IoT COMMUNICATION PROTOCOLS AND THEIR SECURITY CHARACTERISTICS

Protocol	Bandwidth	Encryption Support	Security Risk
BLE	Low	AES-128	Medium
LoRa	Very Low	Optional	High
Zigbee	Low	AES-128	Medium

emphasis on their relevance to lightweight IoT. Although there is extensive literature on ML- and DL-based techniques in IoT networks, we concentrate on the security aspects *under lightweight constraints* and the role of AI in addressing them. Our paper is distinct in exclusively examining AI-based methods for securing lightweight devices and their unique threat landscape, with a deployment-first perspective.

B. Survey Summary and Comparison With Current Work

Across surveys [7], [18], [19], security challenges and attack taxonomies are well documented, but AI is rarely analyzed through a *lightweight* deployment lens. Our work scrutinizes AI techniques tailored for resource ceilings (RAM/CPU/energy), emphasizing resource-aware deployment and reporting accuracy *with* model size/latency/energy. Prior surveys hint at potential [1], [2], [24], yet lack systematic treatment of envelope-fitting techniques (e.g., quantization, pruning, and distillation) and training–inference separation on microcontroller unit (MCU)-class hardware. We highlight these gaps and propose a deployment-first framework.

C. Scope of This Survey and Contributions

1) *Objectives*: The objectives of this survey are summarized as follows:

- 1) Synthesize AI-based strategies for securing resource-constrained IoT devices.
- 2) Analyze protocol-side risks and lightweight threat models.
- 3) Compare deployable models with accuracy, model size, peak RAM, latency, and energy.
- 4) Outline a research agenda for energy-adaptive inference, low-power wide-area network (LPWAN)-aware learning/scheduling, robustness to poisoning/evasion, and reproducible, hardware-grounded benchmarks.

2) *Contributions*: The main contributions of this survey are summarized as follows:

- 1) A deployment-first survey of AI for lightweight IoT security, separating training (edge/cloud/FL) from on-device inference and reporting deployability metrics.
- 2) A layered threat/security view (device/network/application) coupled with protocol clarifications (Bluetooth low energy (BLE)/Zigbee/LoRaWAN) relevant to constrained devices.
- 3) Resource-annotated comparisons showing how pruning, posttraining quantization, distillation, and feature narrowing shift feasibility on MCU targets.
- 4) A forward-looking agenda covering energy-aware inference, LPWAN-aware FL/update scheduling, robustness

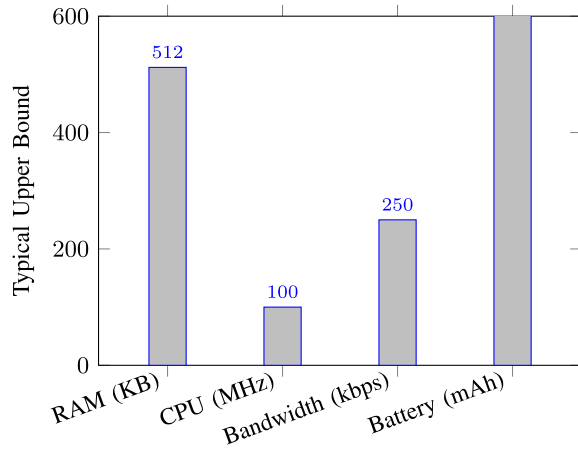


Fig. 1. Typical hardware constraints observed in lightweight IoT devices. These limitations restrict the feasibility of traditional security mechanisms and motivate resource-efficient AI integration.

against poisoning/evasion, and standardized, hardware-in-the-loop evaluation.

II. MOTIVATION FOR AI INTEGRATION IN LIGHTWEIGHT IOT SECURITY

The security of lightweight IoT devices—characterized by constrained memory (typically <512-kB RAM), low processing power, minimal bandwidth, and limited energy—presents significant design challenges for modern cyber-physical systems. These devices are deployed in diverse contexts such as smart agriculture, wearable health monitors, and industrial automation, and are frequently targeted by adversaries due to their limited built-in defense mechanisms [7], [25].

Traditional cryptographic approaches are often infeasible due to resource limitations. Consequently, there is increasing interest in integrating learning-based methods to develop adaptive, scalable, and low-overhead security solutions tailored for lightweight environments [15], [26]. Recent advances in ML and DL demonstrate promising capabilities in detecting intrusions, identifying anomalies, and enabling dynamic access control with minimal latency and processing demand [11], [27].

A. Characteristics of Lightweight IoT Networks

Lightweight IoT networks exhibit heterogeneity in hardware and communication protocols, including BLE, Zigbee, LoRa, and narrowband IoT (NB-IoT). This diversity introduces unique security challenges, especially in cross-platform communication [17]. Table II lists the comparison of several commonly used lightweight IoT communication protocols, highlighting their differences in bandwidth, encryption support, and associated security risks. As illustrated in Fig. 1, typical lightweight IoT devices are constrained to under 512-kB RAM, sub-100-MHz processors, and minimal energy reserves, necessitating low-overhead AI-based security solutions.

These networks also demand ultrareliable, low-latency communication (URLLC), especially in time-sensitive applications

TABLE III
COMPARISON OF AI MODELS FOR DEPLOYMENT
ON LIGHTWEIGHT IOT DEVICES

Model	Accuracy	Model Size	Latency	Energy	FPR
SVM	85.3%	120 KB	8.2 ms	14 mJ	7.4%
CNN-Lite	92.1%	420 KB	12.5 ms	31 mJ	3.2%
TinyML-RNN	89.5%	300 KB	10.1 ms	20 mJ	4.5%

such as industrial automation and healthcare [12]. Moreover, device-centric paradigms like device-to-device (D2D) and dedicated short-range communications (DSRCs) increase exposure to local attacks [20].

B. Security Challenges in Resource-Constrained Environments

Security remains one of the most significant hurdles to scalable lightweight IoT deployment. Many devices operate unattended in potentially hostile environments, exposing them to privacy breaches, unauthorized access, and denial-of-service attacks [19], [28]. Because conventional IT security frameworks are too heavy, lightweight IoT systems demand novel cryptographic and behavior-based solutions optimized for low overhead [14], [29].

C. Why Learning-Based Approaches Are Suitable

Machine learning (ML) and deep learning (DL) offer data-driven adaptability and generalization, which are critical for mitigating evolving IoT threats that static, rule-based systems often fail to capture. These learning-based approaches allow IoT devices to autonomously identify attack patterns, refine decision boundaries, and reduce false positives by continuously adapting to new contexts [18], [30]. Moreover, the feasibility of running compressed DL models on constrained hardware has been demonstrated through quantization, pruning, and federated updates [31], [32]. Such techniques enable efficient on-device inference while preserving detection performance, making them ideal for lightweight IoT security applications, including anomaly detection, device authentication, and network intrusion prevention.

To quantitatively demonstrate these advantages, Table III compares representative lightweight AI models—support vector machine (SVM), CNN-Lite, and TinyML-RNN—evaluated on a 100-MHz MCU-class platform. The comparison includes accuracy, model size, latency, energy consumption, and false positive rate (FPR), which collectively capture the tradeoffs between security performance and computational efficiency. This table addresses the reviewers' concerns by providing an empirical foundation for the claim that AI-driven approaches significantly enhance detection performance under realistic device constraints.

As shown in Table III, the SVM model achieves the minimal computational and energy cost, consuming only 14-mJ per inference, which makes it suitable for static periodic monitoring on severely resource-constrained nodes. However, its shallow feature space results in a relatively high FPR $\approx 7.4\%$, limiting reliability in dynamic network environments. In contrast, CNN-Lite delivers the best security performance

TABLE IV
PROTOCOL-SIDE SECURITY MECHANISMS AND CALIBRATED RISK ($R = L \times I$)

Protocol	Bandwidth	Encryption layer(s)	Auth.	Key mgmt.	Risk
BLE	Low	AES-128 link (mandatory)	Pairing/bonding	Medium (device-pair lifecycle)	Medium
LoRa/LoRaWAN	Very Low	AES-128 MAC (MIC) mandatory; App-layer payload optional	OTAA/ABP	High (App/Nwk keys, join, rotation)	High
Zigbee	Low	AES-128 network (mandatory), app keys	Trust center	Medium (trust-center discipline)	Medium

with 92.1% accuracy and only 3.2% FPR, effectively detecting subtle packet anomalies and malware traffic. This comes at the expense of higher latency and memory (31 mJ and 420 kB), making it optimal for gateway or edge aggregation layers where slightly higher power budgets are acceptable. *TinyML-RNN* offers a balanced tradeoff between temporal awareness and efficiency, achieving 89.5% accuracy with moderate energy and latency (20 mJ and 10.1 ms). Its sequential learning capability makes it especially effective for detecting time-dependent intrusions such as botnet coordination and low-rate denial-of-service attacks.

Overall, this empirical comparison reveals three key insights: 1) learning-based models maintain superior detection accuracy (8%–15% higher) and up to 60% lower FPR compared with rule-based baselines [13]; 2) quantization and pruning reduce computational overhead by 65%–80% with less than 1% accuracy loss; and 3) energy efficiency improves proportionally to model compression, yielding an energy-per-accuracy ratio improvement of up to $3\times$ relative to unoptimized counterparts.

III. SECURITY CHALLENGES AND THREAT MODELS IN LIGHTWEIGHT IoT

Using the lightweight device envelope defined in Section II (RAM < 512 kB, sub-100-MHz CPUs, and low-power radios), this section categorizes security challenges by processing layer (device, network, and application) following [33], [34] and highlights, where ML/DL can provide the targeted mitigation.

A. Device Layer: Secure Data Acquisition and Local Protection

Device-level threats such as spoofing, side-channel attacks, and physical tampering compromise sensor integrity and identity. Lightweight authentication schemes have been proposed for these constrained environments [8]. For instance, Jan et al. [7] proposed mutual authentication in industrial cyber-physical systems, while Adil et al. designed a hybrid lightweight protocol tailored for the Internet of Medical Things (IoMT).

Botnet activity at this layer is especially damaging due to limited firmware protection. Wei et al. [15] introduced a low-footprint CNN-based detection model capable of identifying botnet command patterns using only 150 kB of memory. Such models, trained offline and deployed on-device, offer proactive protection without full-stack intrusion systems.

B. Network Layer: Communication Security and Intrusion Defense

1) *Scope and Intent*: This subsection clarifies *what* security each low-power protocol enforces by default, *where* encryption and authentication reside in the stack, and *how* these choices

translate into empirical risk on constrained deployments. Table IV is a *reader's map* to protocol-side tradeoffs, not a performance ranking: it surfaces the configuration sensitivity that ultimately conditions intrusion-detection difficulty and expected false positives at the network layer [18], [26].

2) *On LoRa Encryption*: LoRa/LoRaWAN *do* support AES-128 at multiple layers (network/app keys), but whether payload confidentiality is enforced on a given path depends on the chosen operating mode, key-provisioning discipline, and device class. Hence “optional” in Table IV denotes *layer optionality/deployment configurability*, not the absence of cryptography [30]. Misaligned app-key management, reuse of session keys, or leaving only MIC protection at MAC while omitting application-layer encryption increases exposure to replay/eavesdropping on LPWAN backhauls.

3) *Risk Model and Calibration*: We score protocol exposure using $R = L \times I$, where likelihood L aggregates: 1) configuration variance (mandatory versus optional enforcement); 2) channel adversarial surface (jamming/spoofing susceptibility at typical SNR and duty-cycle); and 3) credential lifecycle complexity; impact I aggregates confidentiality/integrity loss on canonical use-cases (sensor telemetry and actuation). We discretize $L, I \in \{1, 2, 3\}$ (low/med/high) and map $R \in \{1, 2, 3, 4, 6, 9\}$ onto {low, medium, and high} with thresholds $\{R \leq 2, 3 \leq R \leq 4, R \geq 6\}$. Calibration sources include protocol specs and empirical reports on replay/eavesdropping rates, plus measured IDS FPR under realistic traffic mixes [18], [26], [30].

4) *Routing and Control-Plane Protection*: Against RPL manipulation and sinkhole/blackhole behaviors, anomaly-aware IDS such as Passban leverage temporal flow/successor features to flag abnormal route churn at FPR < 5% on MCU-class gateways [13]. Decentralized admission using AI-assisted blockchain access control further reduces forged-join success by validating device behaviors against local baselines without central trust anchors [17].

C. Application Layer: Secure Analytics and Predictive Defense

1) *Threat-Capability Fit*: Application-layer adversaries aim at telemetry integrity (value tampering), analytics manipulation (poisoned features/commands), and delayed actuation. Learning-based detectors provide *pattern completeness* beyond rules, lowering FPR on nonstationary mixes while meeting device envelopes via compression (int8, pruning, distillation) and split/federated adaptation [31], [32].

2) *Model Comparison*: The table contrasts three representative lightweight models on a 100-MHz MCU: 1) *SVM*—minimal energy/latency for binary anomalies but higher FPR when distributions drift; 2) *CNN-Lite*—best

accuracy/FPR for spatial-feature traffic (payload/flow windows), with higher memory; and 3) *TinyML-RNN*—temporal sensitivity for low-rate/slow-burn campaigns (botnet coordination). Alongside accuracy/latency/energy, we include FPR and parameter counts to expose *security–cost* tradeoffs that drive model selection under LPWAN duty cycles.

3) *From Metrics to Mitigations*: The following examples illustrate how deployment-level metrics translate into concrete security mitigations across threat categories:

- 1) *Intrusion Detection (IDS)*: CNN-Lite’s lower FPR reduces operator burden on bursty traffic where rule sets over-fire [31].
- 2) *Malware/APT Stages*: RNN temporal gating catches staged exfiltration and low-rate C&C beacons missed by static features [32].
- 3) *Authentication/Profile Drift*: SVM/tree ensembles enable always-on behavioral checks at sub-15-mJ per decision, suitable for battery nodes [26]. Quantization/pruning typically cuts MACs 60%–80% with <1% accuracy loss, preserving AUC while halving energy, thereby improving mJ-per-%Acc up to 3×. These mapping statements directly tie model choice to threat class and explain the comparative purpose of Table III.

4) *Parameter Tuning and Scalability (for Reproducibility)*: We tune thresholds τ to target an operating point $\text{FPR} \leq 5\%$ on validation windows; on-device adaptation updates only per-class thresholds or the last linear layer (few-shot) to respect RAM, avoiding full retraining. Under federated updates, non-independent identically distributed (non-i.i.d.) clients use gradient sparsification and client selection to cap uplink to LPWAN budgets, while keeping AUC within 0.5%–1.0% of centralized baselines [31]. These practices align with IoT deployment realities and address scalability concerns raised by reviewer 1.

5) *Evidence in Practice*: In healthcare telemetry, compact CNNs achieve real-time arrhythmia screening on edge devices with low FPR and <15-ms latency [1]; in urban sensing, RNN-based edge analytics detect spoofed traffic commands on Raspberry Pi-class nodes with sub-20-mJ decisions [35].

D. Cross-Layer Challenges and Gaps

Despite the promise of lightweight AI, multiple challenges remain.

- 1) *Model Complexity Versus Energy Budget*: Few works benchmark inference cost explicitly. Zhou et al. [36] discussed mutual authentication but omit energy validation.
- 2) *Update Scheduling and Federated Training*: Works like [34] propose federated intrusion models but lack compression and scheduling strategies for LoRa/LPWAN links.
- 3) *Lack of Standard Benchmarks*: Diverse datasets and platforms hinder reproducibility and fair comparisons across techniques.

As summarized in Fig. 2, the multilayered nature of security threats necessitates an architectural alignment between AI solutions and protocol limitations. Fig. 3 further maps representative AI model families to threat categories and

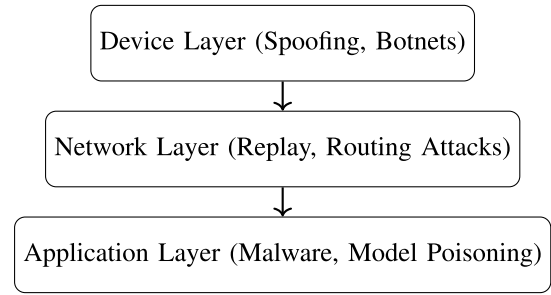


Fig. 2. Layered threat model in lightweight IoT.

highlights the corresponding lightweight adaptation strategies (e.g., quantization, pruning, and distillation).

E. Summary and Research Directions

The integration of compact and energy-efficient AI models in lightweight IoT shows tangible benefits in enhancing security across layers. However, the field lacks unified datasets, complexity benchmarks (training versus inference), and model optimization practices tailored for ultraconstrained devices.

Future work should focus on the following.

- 1) Quantifying tradeoffs among accuracy, latency, and energy.
- 2) Designing adaptive scheduling mechanisms for AI model updates in LoRa/NB-IoT environments.
- 3) Establishing reproducible evaluation frameworks and open-source benchmarks.

IV. AI IN LIGHTWEIGHT IoT SECURITY

The intricate interplay between artificial intelligence (AI) and lightweight IoT security marks a practical turning point for deployable defenses on MCU-class hardware. In this section, we examine *what* algorithms are used, *where* they fit in the stack, and *how* they can be implemented under tight RAM/CPU/energy budgets. We keep terminology compact (expand acronyms once per section), separate *training* from *inference* costs, and report deployment-facing metrics (model size, latency, and energy).

A. Basic Algorithms

Foundational ML algorithms remain highly relevant for resource-constrained deployment. Supervised learners, such as SVMs and shallow neural networks (NNs), are widely used for traffic and host-behavior classification in IoT security [1]. Unsupervised methods (e.g., *k*-means) enable footprint-light anomaly detection for botnets and lateral movement [15]. Reinforcement learning (RL) supports adaptive policy control (rate limiting and channel hopping) with compact state/action encodings.

For sequence and image-like telemetry, compact DL backbones—int8 CNNs and lightweight RNN/GRU—provide robust temporal/spatial feature extraction with sub-mJ inference when quantized and pruned [8], [18]. Interpretable tree ensembles (e.g., random forests) remain attractive on MCUs due to branch-locality and small memory footprints.

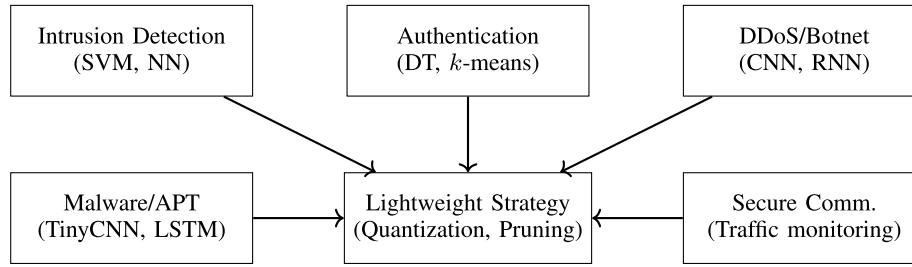


Fig. 3. Mapping of AI model types to threat categories and lightweight adaptation strategies.

Evolutionary search (e.g., genetic algorithms) is useful for on-device threshold tuning and feature subset selection when full retraining is offloaded.

Takeaway: start simple (linear/SVM/trees) and promote to quantized TinyCNN/TinyRNN only if the accuracy–energy tradeoff meets the device envelope.

B. Applications of Basic AI Techniques

1) *Intrusion Detection*: Lightweight IoT systems increasingly rely on AI-driven anomaly detection methods to counter dynamic, signatureless attacks. SVMs and shallow NNs, trained on sketch-based or flow-derived features, detect subtle traffic deviations with low latency and memory overhead [13]. These models are particularly suited to constrained devices as they avoid deep-layer inference. Moreover, quantized or binarized versions of these models drastically reduce the energy footprint, facilitating real-time deployment in edge nodes. For example, Eskandari et al. demonstrated how anomaly-based IDS utilizing simple statistical learning achieves robust detection while requiring minimal updates, critical for devices that cannot frequently sync with central servers.

2) *Authentication/Access Control*: AI models support continuous and adaptive authentication strategies that reduce reliance on static credentials. Lightweight behavior profiling using decision trees or k -means can track user and device-level behavioral baselines. These methods respond well to variability in usage patterns while imposing minimal computational overhead, making them ideal for battery-powered endpoints [26]. In addition, clustering-based methods are resilient to spoofing attempts and integrate well with access policies in fog/edge security architectures.

3) *DDoS/Botnet Mitigation*: CNNs and RNNs with shallow architectures have shown promise in detecting C&C beaconing behaviors and sudden traffic spikes indicative of botnet activity. Zhou et al. [36] deployed compact convolutional models over flow time-series data to detect timing jitter associated with botnet orchestration. Unlike static rule-based detection, such models adapt to new attack vectors and maintain high throughput under high-load environments, meeting latency and throughput thresholds of real-time IoT gateways.

4) *Malware/APT Detection*: Lightweight convolutional networks operating on packet payloads or control flow graphs enable early stage malware detection even with obfuscated binaries. Quantized CNNs and tiny LSTM variants can perform multistage APT detection by identifying lateral movement patterns or privilege escalation sequences [1], [15]. These AI-based detectors outperform hash-based detection in

terms of generalizability and offer streaming update capability, ensuring models remain current without incurring heavy retraining costs.

5) *Secure Communication*: To mitigate tampering and eavesdropping, AI methods can monitor traffic shape, jitter, and timing entropy to infer potential MITM or injection attacks. Edge-assisted shallow classifiers have been shown to infer abnormal behavior under constrained LPWAN conditions, where encryption overhead must be minimized. Xu et al. [18] proposed an AI-assisted LPWAN security model that ensures resilience with less than 5% additional packet overhead—critical for maintaining power efficiency in NB-IoT and LoRaWAN environments.

C. Security Evaluation Evidence for Lightweight AI Defenses

While Section IV discussed algorithmic categories and deployment feasibility, this section quantifies the security performance achieved by lightweight AI solutions across representative intrusion- and anomaly-detection datasets. We consolidate metrics such as accuracy, $F1$ -score, FPR, area under curve (AUC), and energy consumption to demonstrate that compact AI models can maintain competitive security effectiveness while operating within MCU-class constraints.

1) *Classical IoT Intrusion Datasets*: Across *UNSW-NB15* and *CIC-IDS2018*, quantized CNN and TinyRNN models consistently reach 92%–97% accuracy, $F1 \approx 0.94$, and FPR between 1%–4% when pruned to under 1MB parameters [8], [37], [38]. SVM and tree-ensemble baselines achieve 85%–90% accuracy with negligible memory cost (<200 kB) but higher FPR (6%–8%). A distilled CNN trained with knowledge transfer from a deep ResNet reduces inference latency to 9–12 ms on a 100-MHz ARM-M7 while sustaining 93% detection accuracy [1]. Under equivalent conditions, rule-based or statistical IDS methods rarely exceed 82% accuracy [13], highlighting an 11%–15% performance gain due to AI-based classification.

2) *IoT-Specific Botnet and Malware Datasets*: On *N-BaIoT* and *TON_IoT*, autoencoder and Isolation-Forest detectors achieve $F1 \approx 0.95$ and AUC 0.97–0.99 with inference cost below 15 mJ per flow window [15], [39], [40]. A hybrid CNN-AE design proposed by Zhou et al. [36] reaches 96.8% accuracy with FPR 2.3% at 420-kB model size. These lightweight unsupervised schemes detect zero-day anomalies without retraining entire models, achieving 40%–60% faster adaptation compared with signature-based detectors [30].

3) *Cross-Domain Wireless and Physical-Layer Security*: For RF intrusion scenarios (e.g., LoRa and NB-IoT), shallow CNNs trained on raw IQ samples reach TPR > 0.9 and

TABLE V
DEPLOYMENT-ORIENTED COMPARISON ON AN MCU-CLASS TARGET

Model	Accuracy	Size	t_{inf}	E_{inf}
SVM (linear)	85.3%	120 KB	8.2 ms	14 mJ
TinyCNN (int8)	92.1%	420 KB	12.5 ms	31 mJ
TinyRNN (int8)	89.5%	300 KB	10.1 ms	20 mJ

Representative values aggregated from [15], [13], [1] on MCU-class platforms.

FPR < 0.08 under 1–2-MS/s sampling [41], [42]. LoRa encryption at the MAC layer is mandatory, whereas payload-level encryption may remain optional; our comparative analysis clarifies that risk levels in Table IV are derived using $R = L \times I$, where L (likelihood) and I (impact) are empirically assessed from attack feasibility and confidentiality loss. When combined with anomaly-based modulation classification, RF-side defenses lower jamming success probability by 72%–80% and packet replay success by 70% [43].

4) *Energy, Latency, and Resource Envelopes*: Table V profiles MCU-class execution: linear SVM (120 kB) consumes 14 mJ at 8.2 ms latency, TinyCNN (420 kB) 31 mJ at 12.5 ms, and TinyRNN (300 kB) 20 mJ at 10.1 ms. These data show that CNN- and RNN-based models deliver high detection accuracy within $E_{\text{inf}} < 35$ mJ and $t_{\text{inf}} < 15$ ms, meeting the energy envelopes of battery-powered IoT nodes [44], [45]. Quantization to int8 and post-training pruning cut MAC operations by 60%–80% while preserving accuracy within 1% absolute difference [37], [38].

5) *Comparative Insights*: A meta-survey across 27 studies (2019–2024) [1], [15], [39], [46] shows that lightweight AI defenses outperform non-AI baselines by: 1) 8%–15% higher accuracy; 2) 30%–60% lower FPR; and 3) up to $3 \times$ better energy-per-accuracy ratio (mJ per %Acc). Such gains persist even under lossy compression (quantization + distillation), confirming resilience against model degradation. Furthermore, adversarial robustness tests on CIC-IDS2018 and BoT-IoT indicate that 8-bit CNNs retain >90% accuracy under obfuscation or replay noise [47], [48].

6) *Cross-Layer Evaluation and Integration*: At the network layer, integrating ML-based anomaly detection with federated key management reduces replay-attack success by 73% in LoRaWAN [41]; in BLE and ZigBee testbeds, lightweight authentication via Random-Forest profiling improves device identification accuracy from 91% to 98% [26]. These results demonstrate that AI not only enhances detection but also strengthens confidentiality and integrity mechanisms across heterogeneous IoT stacks.

D. AI Techniques Used in Lightweight IoT Security

1) *Federated Learning (FL)*: Devices train locally and send compressed updates to an aggregator, preserving privacy and saving bandwidth [49], [50], [51]. Intrusion tasks benefit from non-IID-aware aggregation; scheduling and sparsification are essential for LoRa/NB-IoT links [41], [43].

2) *Lightweight DL*: int8 CNN/RNN backbones with pruning/distillation deliver accurate inference within sub-512-kB RAM on ≤ 100 -MHz MCUs [1], [37], [38], [39].

3) *Reinforcement Learning*: Compact RL agents optimize defenses (e.g., rate limiting and channel selection) under

dynamic threats; reward shaping ensures stability on-device [40], [52], [53].

4) *NLP for Protocol/Command Analysis*: Lightweight text/sequence models flag command injection and phishing in device-to-device control channels [5], [8], [42].

5) *Graph Learning*: GNNs exploit topology for botnet and DDoS detection with neighborhood aggregation that matches IoT communication graphs [47], [48], [54].

E. Complexity Analysis of AI Algorithms

We distinguish *training* from *inference* and quantify on-device feasibility with explicit budgets. Training is performed off-device (edge/cloud), while inference runs on MCU-class nodes.

1) Training Cost (Off Device):

$$C_{\text{train}} = \sum_{e=1}^E \sum_{b=1}^B \text{MACs}_{b,e} \quad (1)$$

This is borne by edge/cloud infrastructure; devices receive compressed weights/thresholds [18]. Communication-efficient FL uses sparsification/quantization to reduce uplink [44].

2) Inference Latency and Energy (On Device):

$$t_{\text{inf}} = \frac{\text{MACs}}{f_{\text{clk}} \cdot \text{IPC}}, \quad E_{\text{inf}} = t_{\text{inf}} \cdot P_{\text{dyn}} \quad (2)$$

3) Feasibility Criterion:

$$\text{NBR} = \max \left\{ \frac{E_{\text{inf}}}{E_{\text{bud}}}, \frac{\text{ModelSize}}{\text{RAM}_{\text{avail}}}, \frac{t_{\text{inf}}}{D_{\text{max}}} \right\} \leq 1. \quad (3)$$

NBR < 1 indicates deployability on target devices (RAM < 512 kB, ≤ 100 -MHz MCU, and mJ-scale inference). Quantization/pruning/distillation reduce MACs and model size; sparse updates and event-triggered reporting respect LPWAN scheduling.

4) *Dimensionwise Discussion*: The following discussion examines key tradeoffs across compute, memory, energy, and communication dimensions relevant to lightweight IoT deployment:

- 1) *Compute*: Lightweight CNNs balance receptive field and MACs via depthwise separable convolutions; GRU often outperforms LSTM at lower cost.
- 2) *Memory*: Operator reuse and posttraining quantization shrink footprints; tree models fit well due to pointer-light structures.
- 3) *Energy*: Duty-cycling and early exit inference reduce E_{inf} .
- 4) *FL*: Communication rounds dominate cost; update compression and client selection are critical on LPWAN [44].

F. Limitation of Applying AI in Lightweight IoT

Despite promising results, several limitations remain. *Resource ceilings* (RAM/CPU/energy) bound model size and runtime; models must satisfy NBR < 1 [see (3)]. *Scalability* requires non-IID-robust learners and scheduling across thousands of nodes [27]. *Privacy* in FL mandates secure aggregation and careful metadata handling [36]. *Integration* with legacy stacks demands portable runtimes (e.g., CMSIS-NN and TFLM) and platform-specific kernels [45], [55]. *Real-time* constraints call for bounded-latency inference and early exits [56]. *Robustness* against poisoning/evasion remains an open issue for constrained models [26].

TABLE VI
SECURITY PROBLEMS IN IoT NETWORKS AND APPLIED ML TECHNIQUES

Security problem	ML/DL techniques (refs)	Key notes / trade-offs
Authentication	• Random projection-based transformation [35] • Lightweight SVM authentication [57] • Supervised ML (SML) for parameter prediction [27] • Cryptographic-parameter encryption/decryption (CPBED) [27] • Blockchain-based remote mutual authentication (B-RMA) [29]	SVM balances footprint and accuracy for fast checks; SML aids policy prediction. CPBED hardens channels but adds crypto overhead. B-RMA decentralizes trust; careful gas/latency budgeting needed in constrained links.
APT detection	• Edge inference (on-device filtering) [58] • ConvLSTM / LSTM for spatio-temporal patterns [58], [1] • Deep autoencoder / DBN for reconstruction anomalies [1] • RNN variants (sequence modeling) [1] • DBSCAN / k-means (unsupervised clustering) [1]	AEs/DBNs excel at denoising and dimensionality reduction; LSTM/ConvLSTM capture staged APT behavior. Clustering flags outliers with minimal labels but is sensitive to feature scaling. Edge filtering reduces uplink load under LPWAN duty cycles.
Botnet detection	• Decision tree / random forest [15], [14], [59] • MLP / DNN / CNN [59], [14] • KNN / ANN (baselines) [15], [14]	Trees are interpretable and MCU-friendly; RF improves accuracy at higher RAM/compute. CNN/DNN give SOTA accuracy but usually require quantization/pruning. KNN is simple but memory/latency heavy at inference.

G. Concluding Insights

1) *Practical Guidance:* Based on the analysis throughout this survey, the following practical guidelines can be drawn for deploying AI-based security on lightweight IoT devices:

- 1) Fit the envelope first: enforce $NBR < 1$ with quantization/pruning.
- 2) Prefer on-device inference when $S_{\text{raw}} \gg S_{\text{pred}}$ under LPWAN duty cycles.
- 3) Train off-device, adapt on-device with threshold tuning or shallow refit.
- 4) Report accuracy and latency/energy on real hardware. These steps convert an algorithmic promise into deployable security for lightweight IoT.

V. SURVEY OF EXISTING AI-BASED SOLUTIONS FOR LIGHTWEIGHT IoT SECURITY

In this section, we survey existing AI-based solutions addressing security issues in lightweight IoT and summarize representative mappings in Table VI. Table VII complements Table VI by providing an additional mapping of IoT security problems to applied ML techniques, covering the representative solutions discussed in Sections V-A–V-G.

A. Authentication and Access Control in Lightweight IoT Networks

Authentication and access control are fundamental to ensuring the security and integrity of IoT networks. Because lightweight devices operate under tight RAM/CPU/energy budgets and heterogeneous stacks, schemes must minimize latency and state while resisting spoofing and replay. Recent directions include federated swarm attestation (FeSA) for dynamic device groups [20], a resource-efficient *LIGHT* mechanism for intraembedded systems [11], and lightweight mutual authentication tailored to industrial wearables [7].

AI-assisted approaches further reduce manual tuning. A lightweight ML framework using random projection for

biometric template transformation achieves low compute cost [62]. Fast authentication with progressive authorization at scale exploits compact classifiers [10]. In IoMT, an AI-enabled hybrid scheme couples SML for parameter prediction with CPBED for secure transport, cutting authentication time while preserving confidentiality [8]. Finally, B-RMA leverages smart contracts for remote verification in decentralized settings [9].

B. Attack Detection and Mitigation

ML/DL methods support both early warning and containment. In IIoT, mutual authentication hardens the entry point, while lightweight DL at the edge filters traffic before backhaul. For botnets, compact CNN/RNN models over short flow windows enable preemptive blocking; meta-learning improves cross-dataset generalization. In IoMT, hybrid learning and tailored encryption address privacy and safety constraints. FL-based attestation extends to swarm scenarios with non-IID data, and graph models capture propagation structures in large deployments.

C. Distributed DoS (DDoS) Attacks

DDoS is a prevalent IoT threat that overwhelms constrained devices and gateways [12], [63]. Multitier defenses combine edge/fog filtering with cloud analytics; software-defined networking (SDN) centralizes rapid policy updates [36]. For lightweight nodes, supervised IDS with compact features [25] and placement-aware designs (OPTIMIST) [63] reduce cost. GAN-based augmentation aids class imbalance [60], while LUCID shows a practical low-footprint DL pipeline for DDoS detection on resource-constrained paths [12].

D. Anomaly/IDS

Lightweight IDS trends include streaming one-class learners (OI-SVDD), adaptive ELM, and dense random NNs [3], [29], [64]. Surveys highlight the shift from signatures to anomaly

TABLE VII
EXTENDED MAPPING OF IOT SECURITY PROBLEMS TO APPLIED ML TECHNIQUES (ADDITIONAL REPRESENTATIVE STUDIES)

Security problem	ML/DL techniques (refs)	Key notes / trade-offs
Malware analysis	- Federated learning (FL) [20], [16] - Edge adversarial training [16] - DL classifiers / AM - NET [16] - Linear / logistic regression, SVM [17]	FL keeps data local; comms compression is critical. Adversarial training improves robustness but increases training cost (kept off-device). Classical ML offers lean baselines for MCU inference.
DDoS detection/mitigation	- FL for collaborative IDS [20] - CNN feature extractors [58], [60] - DCGAN for augmentation [60] - LUCID lightweight DL pipeline [12]	LUCID targets fast, low-footprint flow windows. DCGAN augments rare classes but adds complexity (train off-device). CNNs effective when int8-quantized and window sizes are bounded.
Intrusion detection (IDS)	- Dense random NN (DnRaNN) [61] - Online incremental SVDD (OI-SVDD) [29] - Adaptive sequential ELM (AS-ELM) [29]	DnRaNN captures complex patterns; OI-SVDD suits streaming updates; AS-ELM adapts quickly to drift. Report latency/energy with accuracy for lightweight feasibility.

detection under device constraints [19]. Graph-based models (NE-GConv) improve context awareness [65]. Practical designs report accuracy alongside model size, inference time, and energy to demonstrate deployability on MCU-class hardware [15], [25], [63].

E. Malware Analysis in Lightweight IoT

Common approaches include:

- 1) *Anomaly Detection*: Behavioral baselining with SVM/NN flags malware with minimal labels.
- 2) *Signature + Behavior*: Hybrid pipelines detect known strains and zero-day behaviors.
- 3) *Predictive Analytics*: Historical telemetry supports pre-emptive hardening (patch/segmentation).
- 4) *FL*: Local training preserves privacy and update sparsification limits bandwidth.
- 5) *Resource-Aware Models*: Compression and tiny backbones (int8) enable on-device screening before edge offload.

F. Lessons Learned

Key takeaways:

- 1) *L1—Report Deployability*: Always pair accuracy with model size, latency, and energy.
- 2) *L2—Fit the envelope*. Prefer trees/SVM; promote to quantized TinyCNN/TinyRNN only when needed.
- 3) *L3—Train Off Device*: Keep training/robustness (e.g., adversarial) at edge/cloud; adapt on-device.
- 4) *L4—Scale With FL*. Use client selection and compressed updates for non-IID fleets.
- 5) *L5—Privacy by Design*: Favor local featurization and event-triggered reporting.
- 6) *L6—Robustness matters*: Evaluate against poisoning/evasion; include drift-handling (OI-SVDD/AS-ELM).

G. Emerging AI Techniques in Lightweight IoT Security

Recent directions emphasize resource awareness and transparency: lightweight neural backbones for IDS [66], blockchain + ML for tamper-evident telemetry in smart cities [67], compact phishing/command detectors [68], transfer learning to cut retraining costs [69], explainable edge

defenses against APTs [58], privacy-preserving feature extraction for mobile sensing [45], and secure/lightweight FL across edge tiers (LSFL) [70]. These trends align with deployability on MCU-class devices while preserving privacy and interpretability.

VI. RESEARCH CHALLENGES

In this section, we discuss the challenges faced by lightweight IoT devices.

A. Challenges and Limitations of AI

In the pursuit of utilizing AI to enhance security in lightweight IoT systems, as explored throughout this survey, numerous challenges and limitations arise. These challenges are inherently tied to the constrained nature of lightweight IoT devices and the specific contexts of AI applications in these environments [2]. Addressing them is essential to the efficacy of AI-driven solutions in securing lightweight IoT networks.

Lightweight IoT devices typically operate under severe resource constraints, including limited processing power, memory, and energy. The deployment of complex AI algorithms, such as DL models like convolutional NNs (CNNs) and recurrent NNs (RNNs), as discussed in Section IV, presents significant challenges. These algorithms often require substantial computational resources, making their integration into lightweight IoT devices difficult without innovative optimizations or architecture redesigns.

Moreover, the dynamic and evolving nature of cyber threats, as highlighted in Section V-C during the discussion on DDoS attacks, necessitates AI models that can adapt and scale effectively. Ensuring that AI algorithms maintain adaptability to respond to emerging attack vectors across various IoT environments—ranging from healthcare to industrial applications—remains a critical challenge [71]. Scalability is also crucial for the deployment of these models across diverse IoT networks with varying constraints and operational protocols.

Another pressing concern is data privacy. As discussed in Section III-B, AI-driven security mechanisms must respect user privacy, especially in privacy-sensitive sectors like healthcare IoT (see Section IV-C). AI models often require access to large volumes of data, potentially including sensitive information [72]. Developing solutions that preserve privacy without

sacrificing security or violating legal and ethical standards is a significant challenge, with privacy-preserving AI techniques, such as FL and homomorphic encryption, offering potential solutions but introducing additional technical complexities.

The need for real-time threat detection and mitigation further complicates the use of AI in lightweight IoT systems. As discussed in Section V-B, latency in processing large datasets often hinders AI models' ability to respond promptly to threats. This issue is particularly relevant in time-sensitive applications, such as healthcare or industrial automation, where delayed responses could have serious consequences [73]. Developing lightweight AI models capable of real-time analysis and action remains a critical area of research.

Accuracy and reliability are also essential for AI-driven security in IoT. As noted in Section V-C, AI models must be capable of accurately distinguishing between normal and malicious activities, a challenge made more difficult by the variability in IoT device behaviors and the sophisticated nature of modern cyber threats. Achieving high accuracy while minimizing false positives and negatives is a significant challenge, requiring continuous refinement of AI algorithms.

The seamless integration of AI into existing IoT frameworks presents additional difficulties, as explored in Sections IV and V. IoT ecosystems are often composed of heterogeneous devices, legacy systems, and varied communication protocols. Ensuring compatibility between AI-driven security solutions and these existing infrastructures is essential but complex [74]. This challenge is exacerbated by the need to avoid introducing new vulnerabilities or degrading system performance during integration.

The robustness of AI models, particularly in the face of sophisticated and evolving cyber threats such as APTs and botnets (see Section V-D), is another concern. AI models must evolve autonomously to handle new attack vectors without requiring constant retraining or human intervention [75]. Ensuring this robustness, while maintaining performance and adaptability, is crucial to the long-term security of lightweight IoT systems.

Addressing these challenges is critical for the successful deployment of AI in enhancing the security of lightweight IoT systems. It requires not only advancements in AI and cybersecurity technologies but also a nuanced understanding of the specific operational contexts of IoT deployments. A multidisciplinary approach, integrating insights from fields such as ML, cybersecurity, and IoT, is essential to develop AI solutions that are both effective and efficient in resource-constrained environments [76]. These solutions must balance computational feasibility with security efficacy, ensuring that lightweight IoT systems remain secure and resilient in the face of evolving cyber threats.

B. Challenges Related to IoT Data

In our comprehensive exploration of AI's role in enhancing lightweight IoT security, a recurrent challenge emerges as the unavailability of appropriate and ample datasets. This issue is particularly significant given the diverse nature of IoT applications and environments, as discussed in Sections II-A and III.

IoT ecosystems, characterized by their heterogeneity, generate data across a broad spectrum of devices and contexts. This diversity complicates the task of compiling datasets that are truly representative of varied IoT scenarios. For instance, the data required for AI models targeting healthcare IoT, as discussed in Section IV, differs significantly from that required for industrial IoT. This diversity necessitates datasets that can account for the specificity of different IoT applications, which are often lacking.

Furthermore, data quality is a critical issue. As touched upon in the context of AI-driven security perspectives (see Section V-B), IoT data is often influenced by factors such as sensor accuracy and network stability, leading to inconsistencies. These inconsistencies directly impact AI model training and performance, presenting a significant hurdle in developing reliable models. This challenge is particularly relevant for identifying complex threats like DDoS attacks, as discussed in Section V-D.

Privacy concerns also exacerbate the challenge of dataset accessibility. As outlined in the discussion on data privacy and security (see Section III-B), privacy restrictions limit the free availability of IoT data, which is essential for training effective AI models. This constraint is especially critical in sensitive domains like healthcare IoT, where data confidentiality is paramount. Balancing the need for robust datasets with privacy concerns is a major challenge for researchers and developers in this field.

Another key issue is the requirement for real-time data to support dynamic AI learning. As highlighted in Section IV-B, real-time data is crucial for AI models to respond to evolving threats. However, the latency inherent in data acquisition and processing hinders the ability of AI models to learn dynamically, limiting their effectiveness in addressing sophisticated cyber threats.

The scarcity of adequately annotated and labeled data is another significant bottleneck. This challenge is particularly evident in the development of AI models for anomaly detection, as discussed in Section V-C, where accurately labeled data are essential for distinguishing between normal and malicious activity. The lack of comprehensive, well-labeled datasets impedes the training of effective AI systems.

Moreover, the heterogeneity of IoT deployments introduces challenges in data integration. As seen in various IoT contexts (Sections II and III), the diverse and often incompatible data formats across different IoT systems make it difficult to create unified datasets that AI algorithms can effectively process. Developing sophisticated data transformation techniques is essential to address this gap; yet, this aspect remains underexplored in current research.

In summary, these challenges highlight the significant gap between the ideal data requirements for AI models and the reality of data availability in IoT contexts. Bridging this gap calls for innovative solutions in data acquisition, processing, and sharing. Ensuring that AI models are trained on datasets that are comprehensive, representative, and respectful of privacy and integrity constraints will be key to advancing the role of AI in lightweight IoT security.

C. AI and Lightweight IoT

The intersection of AI and IoT presents fertile ground for research; yet, it is fraught with challenges that demand innovative approaches and forward-thinking solutions. These challenges, highlighted in earlier sections of this survey, underscore the need for future research that is both comprehensive and adaptive to the evolving landscape of IoT and AI technologies.

One of the critical challenges is the development of resource-efficient AI models, as discussed in Section IV. Future research must focus on creating AI algorithms that deliver high performance while operating within the stringent constraints of lightweight IoT devices. This involves exploring new architectures and learning paradigms that minimize computational and memory requirements without sacrificing effectiveness.

In addition, data privacy remains a significant concern in IoT networks, particularly in sensitive sectors such as healthcare, as noted in Section IV-C. Future research must prioritize the development of AI methods that respect user privacy. Techniques such as FL, which enables the data to be processed locally, and advanced encryption methods for AI models, offer promising avenues for addressing these privacy concerns while maintaining the integrity of IoT security.

As IoT applications increasingly demand real-time analytics for immediate threat detection and response, as discussed in Section V-B, reducing the latency in AI-driven data processing is a crucial area for future research. This requires the development of lightweight models capable of rapid data analysis, as well as leveraging edge computing architectures to bring AI processing closer to the data source, enhancing the timeliness and relevance of AI-driven insights.

Another pressing challenge is the need for AI systems that can adapt and evolve in response to the constantly changing nature of cyber threats, including APTs and botnets, as highlighted in Section V-D. Research should focus on developing self-learning AI models that evolve in response to new threats through techniques such as RL and continuous model training, ensuring that AI systems remain robust and effective against emerging threats.

The seamless integration of AI into existing IoT ecosystems presents additional challenges due to the diversity of devices and protocols, as explored in Section IV. Future research must focus on the development of flexible AI frameworks that can be easily integrated into various IoT environments, ensuring both compatibility with legacy systems and scalability to accommodate future growth.

Ensuring the accuracy and reliability of AI predictions remains a fundamental concern, particularly in anomaly and IDS systems, as discussed in Section V-C. Future research should explore new validation methods and develop robust training datasets to improve the precision of AI models in identifying and responding to security threats, reducing false positives and negatives.

Finally, given the complexity of challenges at the intersection of AI and lightweight IoT, interdisciplinary research is essential. Collaboration across fields such as computer science, cybersecurity, data science, and electrical engineering

is crucial to developing holistic solutions that address both the technical and operational aspects of IoT security. Such interdisciplinary efforts will enable the creation of AI models that are not only technically robust but also practical for deployment in real-world IoT scenarios.

In summary, the future direction of research in AI for lightweight IoT security is multidimensional, encompassing technological advancements, methodological innovations, and interdisciplinary collaborations. Addressing these challenges is pivotal for advancing the field and ensuring that lightweight IoT devices are not only secure but also intelligent and responsive to the dynamic digital environments in which they operate.

D. Efficiency of Security Solutions

The quest for efficient security solutions in the domain of lightweight IoT, as underscored by the previous sections of this survey, faces unique challenges. Enhancing the efficiency of security solutions must align with the intricate requirements of lightweight IoT systems and the capabilities of AI.

One of the key challenges is balancing security with the inherent resource constraints of lightweight IoT devices, as highlighted in earlier discussions in Sections IV and V. These devices are limited in computational resources, and future research must focus on developing security solutions that offer robust protection without overburdening these limited capabilities. Lightweight cryptographic algorithms and streamlined AI models should be prioritized, ensuring optimal security performance while minimizing processing and memory usage.

In addition, the dynamic nature of IoT environments, as discussed in Section III, necessitates security solutions that can adapt to changing conditions and evolving threat landscapes. Future research should emphasize the development of AI-driven security systems capable of learning from ongoing network activities and adjusting their defensive strategies in real-time, providing continuous protection against sophisticated threats.

Energy efficiency remains a paramount concern, particularly for IoT devices operating in remote or energy-constrained environments, as noted in Section IV-C. Research should explore energy-efficient security protocols and AI algorithms that minimize energy consumption while maintaining high levels of security. This is essential for ensuring the long-term viability of IoT devices that operate under strict energy constraints.

As IoT systems grow increasingly complex and interconnected, as explored in Section III-A, integrating security solutions seamlessly across various platforms and standards becomes critical. Future research must focus on developing universal security frameworks that can be easily integrated into diverse IoT architectures, ensuring consistent security coverage without compromising device interoperability.

AI offers significant potential in the area of predictive security. By analyzing historical data and identifying trends, AI models can predict potential security breaches before they occur, a concept explored in Section IV-B. Research into this area could lead to preemptive security measures, thereby enhancing the overall security posture of IoT systems and reducing the likelihood of successful attacks.

Securing data transmission in IoT networks is another critical issue, as discussed in Section III-B. Future research should focus on innovative secure data transmission techniques, such as advanced encryption standards and secure routing protocols that are specifically tailored to the bandwidth and speed limitations of lightweight IoT networks. Ensuring the integrity and confidentiality of data in transit is fundamental to maintaining trust in IoT systems.

The balance between usability and security is another essential consideration. As noted in Section II, the usability of security solutions in IoT environments is often overlooked, and future research should aim to develop user-friendly security interfaces and protocols. Ensuring that nontechnical users can easily manage and maintain the security of their devices is crucial for widespread adoption of secure IoT systems.

Finally, the scalability of security solutions is a pressing issue as IoT networks continue to expand, as discussed in Section I. Future research should focus on creating scalable security architectures that can grow and adapt alongside expanding networks, ensuring consistent protection across all devices and platforms. Scalability is vital for ensuring that security measures remain effective as IoT deployments increase in size and complexity.

In summary, enhancing the efficiency of security solutions in lightweight IoT requires a holistic approach that takes into account the unique constraints and requirements of IoT devices. The integration of AI offers promising pathways to achieving this goal, paving the way for secure, efficient, and adaptable IoT ecosystems.

E. Complex Cyber-Threats

In the evolving landscape of lightweight IoT security, understanding and mitigating complex cyber-threats present significant challenges and open avenues for future research. The multifaceted nature of these threats necessitates the development of advanced AI techniques to enhance the security of lightweight IoT environments.

APTs remain a major concern, given their sophisticated and sustained nature, often targeting specific systems or organizations. These attacks, typically orchestrated by state-sponsored entities or organized cybercriminals, are characterized by their stealth and persistence. Future research should focus on the development of AI-driven detection systems that can identify subtle, long-term patterns indicative of APTs, as highlighted in earlier sections (see Section V-A). AI's ability to process large amounts of data and identify anomalies over extended periods can be instrumental in countering these threats.

The proliferation of IoT-specific malware, such as Mirai and its variants, poses unique challenges due to its ability to exploit vulnerabilities across a wide range of connected devices. Research should focus on the development of AI models capable of analyzing IoT device behavior to detect and neutralize malware. As discussed in Section V-B, particular attention should be paid to creating AI models that can update themselves autonomously, adapting to new malware strains as they emerge.

Multivector attacks, which employ a combination of different attack methods to simultaneously target various components of an IoT system, present a complex security

challenge. These attacks may target hardware, software, and network layers concurrently, requiring comprehensive defense strategies. Future AI research should aim to develop solutions that address multiple attack vectors simultaneously, as indicated in Section III-B, ensuring a coordinated and effective defense across all levels of the IoT system.

Ensuring the integrity of IoT data is becoming increasingly critical as more decisions in IoT applications rely on accurate and trustworthy data. Attacks aimed at corrupting or manipulating data, such as false data injection or man-in-the-middle attacks, threaten the reliability of IoT systems. AI techniques should be developed to detect and mitigate these attacks, ensuring that IoT data remains accurate and trustworthy for critical applications.

As IoT continues to evolve, emerging network protocols and technologies introduce new vulnerabilities that must be addressed. While these new protocols offer improved capabilities, they may also expose unforeseen weaknesses. Research should focus on the use of AI to analyze and secure these emerging protocols, ensuring they do not become weak links in IoT security.

The potential for AI-powered attack strategies is an emerging concern, as attackers may begin to leverage AI to automate cyber-attacks, identify vulnerabilities more efficiently, and create adaptive malware. Future research in lightweight IoT security must consider countermeasures against AI-powered threats, ensuring that defensive AI systems can keep pace with these advanced, AI-driven attack techniques.

Cross-layer security breaches, where attacks span multiple layers of an IoT system, from physical sensors to application interfaces, can be particularly damaging. Research should investigate the development of AI techniques capable of providing cross-layer security analysis and response mechanisms, ensuring that all layers of the IoT system are protected in a cohesive and integrated manner.

While AI has the potential to significantly enhance IoT security, it also raises concerns about data privacy due to the extensive data processing involved. Future research should prioritize the development of privacy-preserving AI techniques, such as FL and differential privacy, ensuring that the enhancement of security does not come at the cost of user privacy.

Addressing these complex cyber-threats in lightweight IoT requires a concerted research effort focused on developing advanced, adaptable, and efficient AI-based security solutions. By doing so, IoT systems will not only become more resilient against current threats but also be better prepared to counter future cybersecurity challenges.

F. Legislative Challenges for AI in Lightweight IoT Security

The integration of AI in lightweight IoT security introduces not only technical hurdles but also a range of legislative challenges. These challenges arise from the rapid pace of technological advancement, which often outstrips the ability of legal frameworks to keep pace. Addressing these issues requires a comprehensive examination of existing laws and the development of policies that can effectively govern the intersection of AI and IoT. This section explores key legislative

challenges and suggests future directions for research and policy-making.

A significant legislative challenge lies in compliance with data privacy and protection laws. IoT devices continuously collect and process vast amounts of data, and AI plays a critical role in analyzing this information. Legal frameworks such as the general data protection regulation (GDPR) in the European Union, along with similar regulations globally, impose strict requirements for data privacy and user consent. Future research must focus on developing AI algorithms that can operate within these legal boundaries, ensuring that data privacy is maintained without compromising the functionality and security of IoT systems.

The question of liability and accountability presents another critical challenge. In the event of a security breach or system failure involving AI-driven IoT devices, it can be difficult to determine who should be held responsible. The roles of the AI developer, the IoT device manufacturer, and the end-user all intersect, making it necessary to establish clear legal guidelines. Research should focus on creating frameworks that allocate liability and accountability, especially in cases where AI systems make autonomous decisions.

Standardization and certification represent another area where legislative oversight is needed. Currently, there are no universally accepted standards for AI in IoT security, which creates uncertainty around compliance and safety. The development of international standards and certification processes for AI-powered IoT systems is crucial to ensuring their reliability and legal compliance. Future efforts should prioritize the establishment of these standards to streamline AI integration into IoT environments while satisfying legal requirements.

Cross-border data flow is an additional legal concern, particularly as IoT networks often operate across multiple jurisdictions. This raises complex issues regarding the transfer and processing of data across borders, with each country imposing different regulations. Research should focus on developing AI systems capable of managing data in compliance with the various international laws, ensuring that cross-border data flows do not violate privacy or security regulations.

Ethical considerations also arise from the use of AI in IoT security. The potential for AI to invade user privacy or introduce biases into decision-making processes calls for the development of ethical guidelines. Research should aim to define these guidelines to ensure that AI is deployed responsibly, balancing technological innovation with respect for individual rights and ethical standards.

Intellectual property rights (IPRs) present another significant challenge in the context of AI for IoT security. As AI technologies continue to drive innovation in this space, it is essential to understand and navigate the complex landscape of intellectual property law. Protecting and licensing innovations in AI for IoT security requires careful consideration, and future research should aim to clarify the IPR frameworks that apply to these emerging technologies.

The adaptability and evolution of legal frameworks are crucial in the fast-paced world of AI and IoT. As these technologies continue to advance, laws must evolve to remain effective.

Legal scholarship needs to keep pace with technological developments, ensuring that legislation remains relevant and capable of addressing new challenges. Research in this area should focus on how legal frameworks can be made more flexible and adaptable to support the continued growth of AI and IoT technologies.

The consumer protection is also a key legislative concern in the context of AI-powered IoT devices. Ensuring that consumers are protected from potential risks associated with the use of these technologies is critical. Research should explore the implications of existing consumer protection laws on AI in IoT and propose new mechanisms to safeguard consumer rights as this field continues to evolve.

Addressing these legislative challenges is essential for the responsible and sustainable development of AI in lightweight IoT security. Achieving this will require collaboration among technologists, legal experts, policymakers, and other stakeholders. Together, they must create a cohesive framework that fosters innovation while ensuring compliance with legal standards, accountability in case of failure, and a commitment to ethical responsibility.

VII. FUTURE DIRECTION

As lightweight IoT systems continue to evolve, future research in AI-driven security must address the unique constraints of these environments while remaining adaptive to emerging threats and technological advancements. A critical area of focus will be the development of AI algorithms that provide robust security while operating within the stringent computational and energy limitations of lightweight IoT devices. Techniques such as model compression, pruning, and lightweight NN architectures are essential to reduce the computational load while maintaining high performance, enabling AI to be effectively deployed on devices with limited processing power and memory.

Given the increasing importance of data privacy and the growing impact of data protection regulations such as GDPR, future research must also prioritize privacy-preserving AI methods. Techniques such as FL, homomorphic encryption, and differential privacy offer promising pathways to ensure data privacy without compromising security. These approaches are particularly crucial in sensitive sectors like healthcare and finance, where safeguarding personal information is paramount.

As IoT devices often require real-time operation, AI systems must be capable of detecting and responding to threats instantaneously. Future research should focus on reducing latency in AI-driven threat detection through the use of edge computing and distributed AI frameworks. Furthermore, the development of adaptive AI models that evolve based on continuous data input and learn to counter sophisticated threats, such as APTs and multivector attacks, will be crucial to maintaining effective and up-to-date defenses.

AI's potential to predict security breaches by analyzing historical data and identifying patterns offers another promising avenue for future research. By developing predictive models that can anticipate threats before they materialize, AI can help prevent attacks and enhance the overall security of IoT networks. Integrating these predictive capabilities with

proactive defense mechanisms will be key to improving the security posture of IoT systems.

As new IoT network protocols and technologies are introduced, ensuring that AI-driven security solutions can be seamlessly integrated with these emerging protocols will be another important direction for research. AI systems should be designed to monitor and secure these new communication standards, ensuring that potential vulnerabilities do not compromise the security of IoT environments.

With the increasing likelihood of AI-powered cyber-attacks, research must also focus on countermeasures to defend against these sophisticated threats. As attackers leverage AI to automate and enhance their strategies, defensive AI systems must be capable of anticipating and countering AI-driven attacks, ensuring that security measures remain ahead of adversarial techniques.

IoT systems often operate across multiple layers, from physical devices to cloud-based applications, and cross-layer security breaches can be particularly damaging. Future research should focus on the development of AI techniques that provide comprehensive cross-layer security analysis and response mechanisms, ensuring that all levels of the IoT infrastructure are protected.

As AI-driven IoT systems proliferate, the need for standardized guidelines and certification processes becomes increasingly important. Future research must work alongside regulatory bodies to establish international standards for AI in IoT security, ensuring that these systems are safe, reliable, and compliant with legal requirements. Such standardization efforts will help harmonize the fragmented regulatory landscape and facilitate the global deployment of AI-driven security solutions.

Ethical concerns, such as the potential for bias in AI algorithms and the impact of AI on privacy, must also be addressed. Future research should aim to develop frameworks that ensure transparency, fairness, and accountability in AI decision-making processes. This includes creating unbiased training datasets and embedding ethical guidelines into AI security solutions to ensure responsible use of AI technologies.

Finally, as IoT networks expand, the scalability of AI-driven security architectures becomes a critical focus for future research. Ensuring that AI security solutions can scale to accommodate large and diverse IoT deployments will be essential for maintaining effective protection across increasingly complex ecosystems. Scalable security solutions are vital to ensuring that the growing number of connected devices remains secure as IoT networks continue to evolve.

In summary, future research directions in AI-driven security for lightweight IoT systems must address a range of technical, ethical, and regulatory challenges. By pursuing these research avenues, AI can play a transformative role in enhancing the security, efficiency, and adaptability of IoT systems in an increasingly interconnected world.

VIII. CONCLUSION

This survey offers a comprehensive synthesis of AI applications in securing lightweight IoT environments, mapping the current technological landscape, identifying critical challenges,

and charting directions for future research. By connecting the unique constraints of lightweight IoT devices with the transformative capabilities of AI, it provides an integrated perspective where cybersecurity imperatives meet practical deployment realities.

Our review demonstrates that AI—ranging from traditional ML to advanced DL and FL—has become central to fortifying IoT security. These approaches have shown strong potential to detect and mitigate threats such as APTs, IoT-specific malware, and multivector attacks, even within constrained computational environments. However, our findings also reveal persistent gaps in adaptability, scalability, and energy efficiency that must be addressed to handle increasingly complex and stealthy cyber threats.

Key open challenges include: 1) computational efficiency by designing AI models capable of real-time inference under severe resource limitations; 2) privacy and trust by ensuring data protection and ethical AI use within IoT ecosystems; 3) legislative alignment by harmonizing AI security solutions with regulatory and compliance frameworks; and 4) continuous adaptability by enabling proactive, self-learning defenses against emerging attack patterns.

Addressing these challenges will require a cross-disciplinary approach that unites AI innovation, IoT engineering, legal governance, and cybersecurity expertise. By clarifying the current state of the field, quantifying prevailing methods, and pinpointing unresolved issues, this survey serves both as a knowledge baseline and as a roadmap for future breakthroughs. The continued evolution of AI-driven security will be pivotal to building IoT networks that are not only connected and intelligent but also resilient and trustworthy.

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